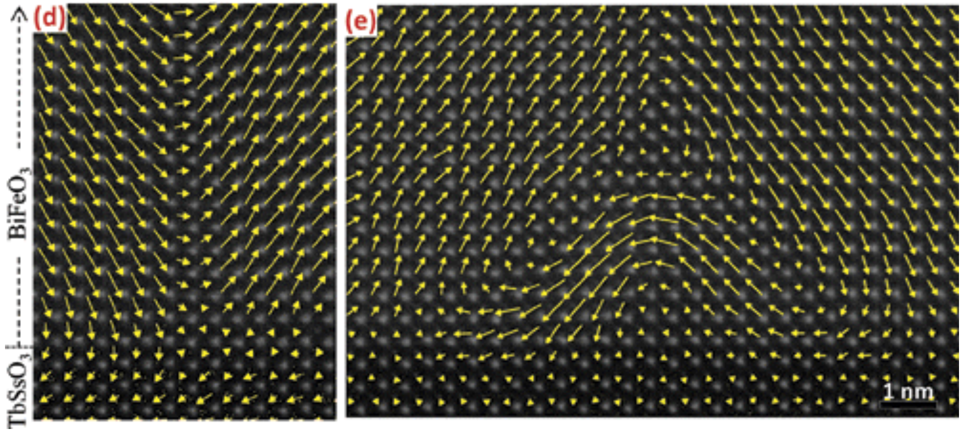
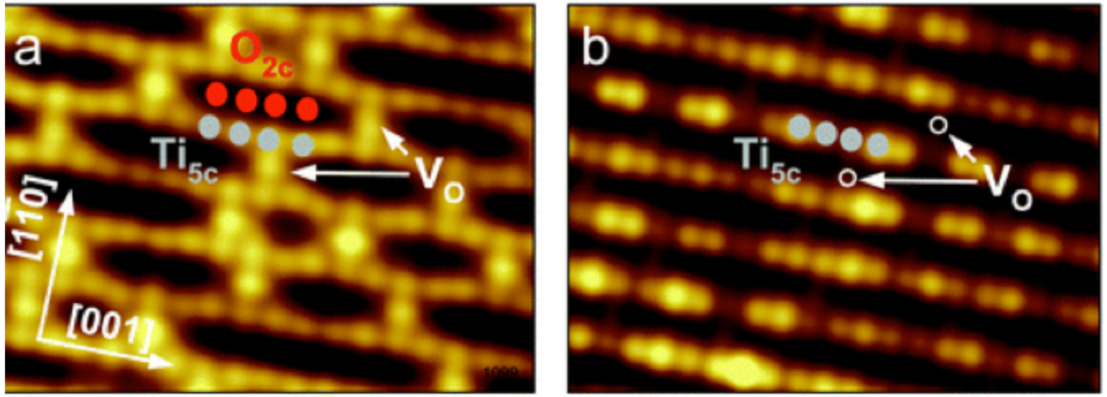
Motivation

Dramatic increase in atomically resolved data from new imaging techniques (microscopy advances)



Closure states in BiFeO3
Nano Lett., **2011**, *11* (2), pp 828

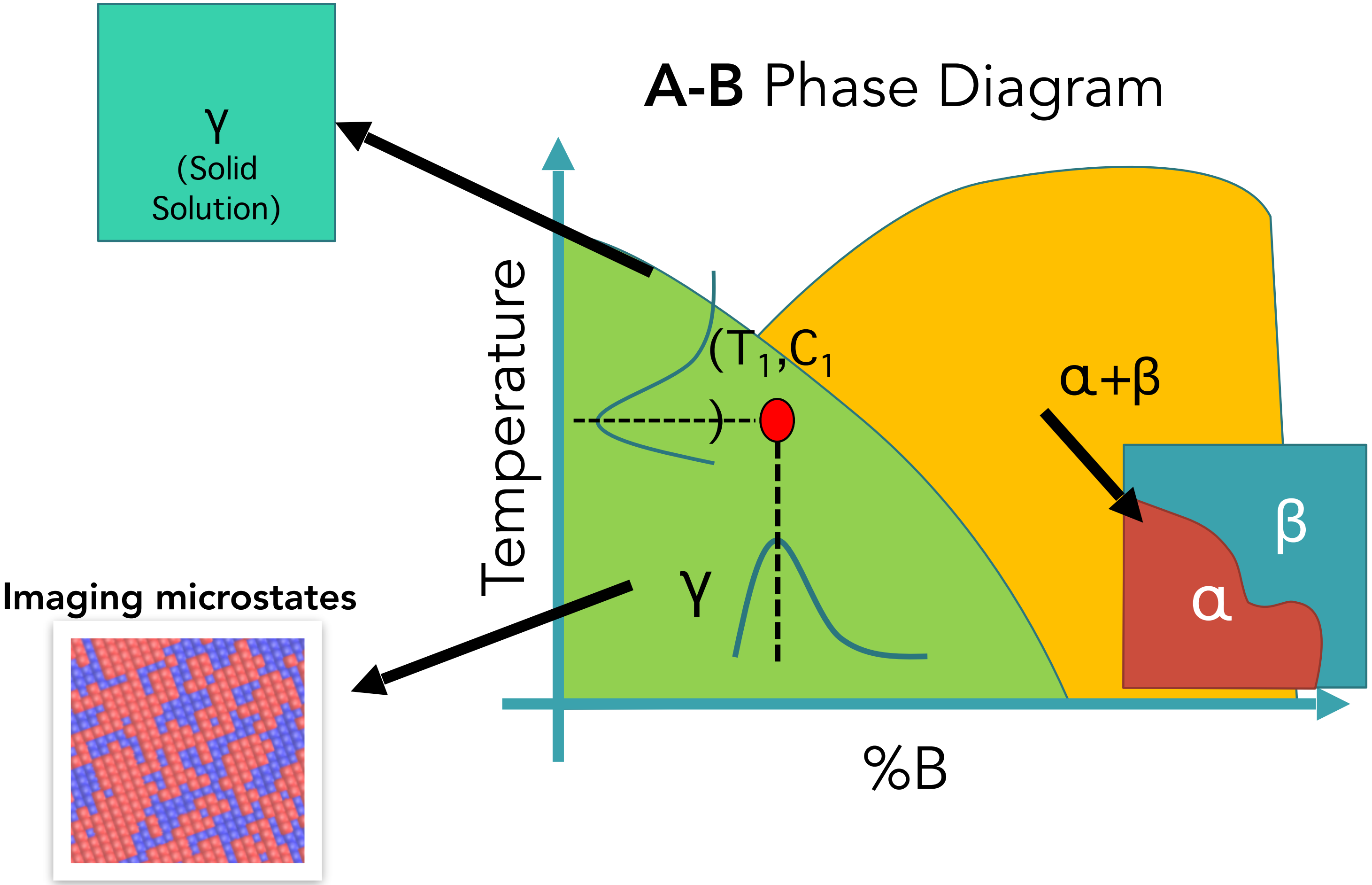


STM (const. height), V_s = +0.7 V, 5x3.5 nm^2
STM (const. height), V_s = -1.2 V

STM rutile TiO_2(110) surface
Chem. Soc. Rev., 2017, **46**, 1772

While image analysis techniques are well developed, only limited amount of transferable materials science knowledge is extracted

What an atomic image really tells us



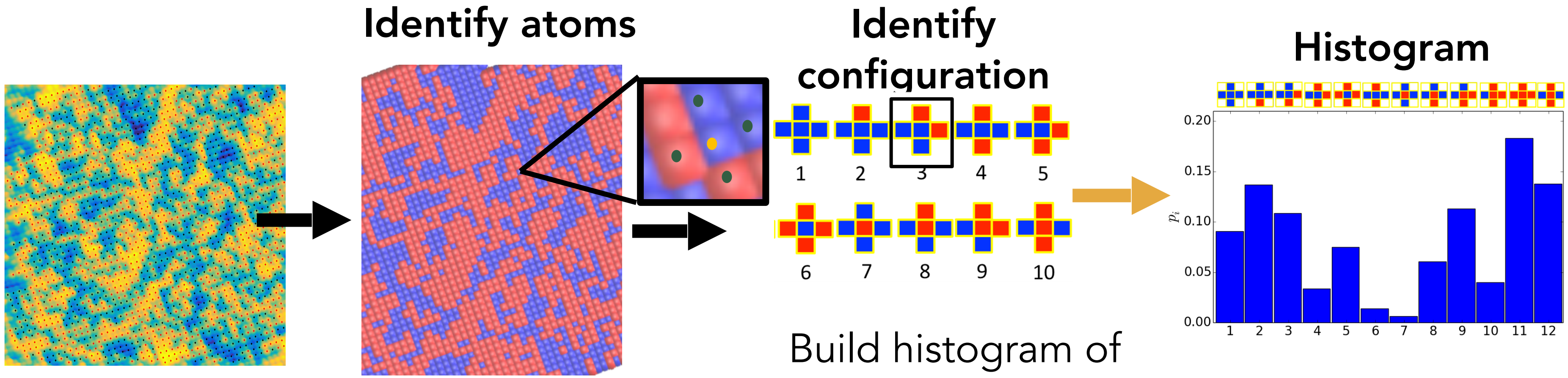
- Response of system to (T,C) is encoded within the imaged fluctuations!
- If we can image this, then can we use it to predict the phase diagram?

What do we need to make it work?

1. We need a generative model (e.g., lattice model (Ising type))
2. We need to compare the right statistics:
   - In the case of atomically resolved images, these will be the atomic configurations
   - Can also be augmented with e.g. pair distribution functions, etc.
3. We need a metric to compare the model to the experimental data, which will correctly capture fluctuations.
   - The appropriate metric in this instance is the statistical distance, which equalizes statistical weights of experimental and statistical data, and enables appropriate comparison
   - This is a measure of distinguishability of two systems (e.g., similar to Kullback-Leibler divergence)

Atomic configurations

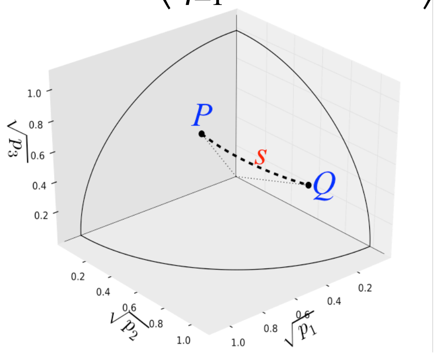
- For atomically resolved imaging, we need to compare the probability of imaging particular atomic configurations

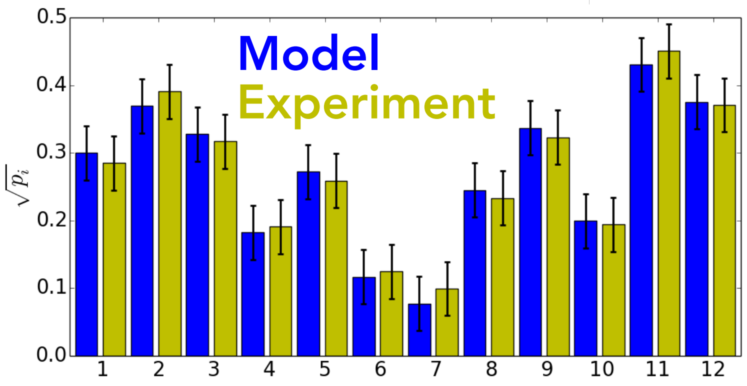


Build histogram of numbers of the configurations (automated)

Statistical Distance Metric

$$s = \arccos\left(\sum_{i=1}^{k} \sqrt{p_i}\sqrt{q_i}\right)$$

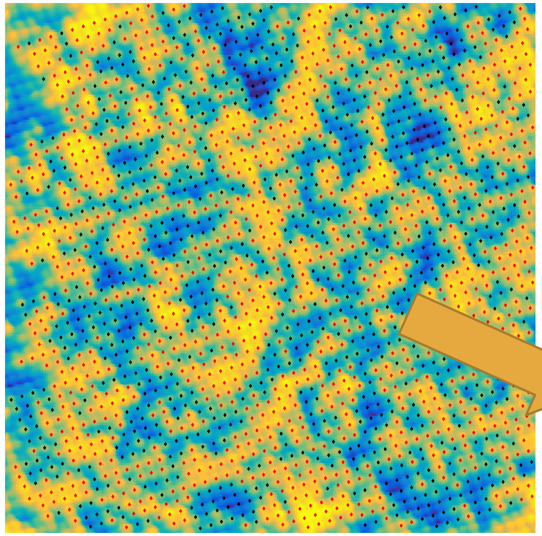


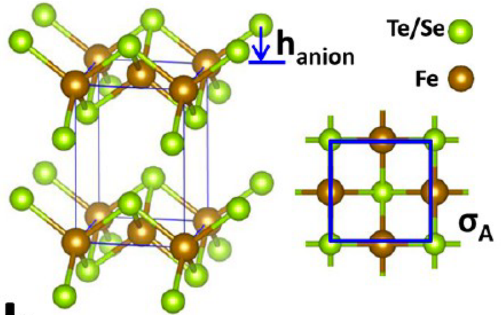


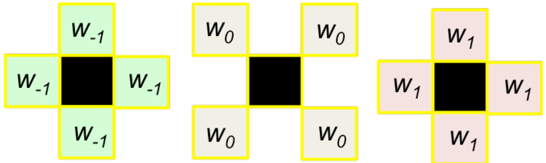
- For the generative model to be predictive, it requires that when we sample from it, those samples are indistinguishable from the target system
- For thermodynamic systems, the distinguishability is expressed in terms of statistical distance, which measures the shortest distance in the probability space
- This metric also equalizes statistical weights of the experimental and simulated data
- Correctly captures fluctuations: essential for predictive abilities

L. Vlcek, A. Chialvo, J. Chem. Phys. 143, 144110 (2015)

Stochastic model matching



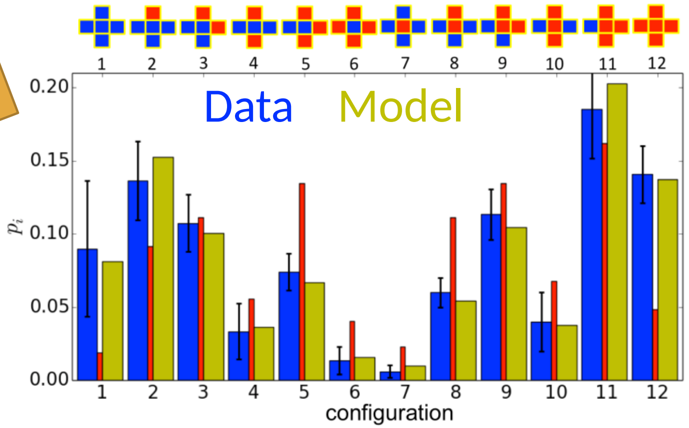


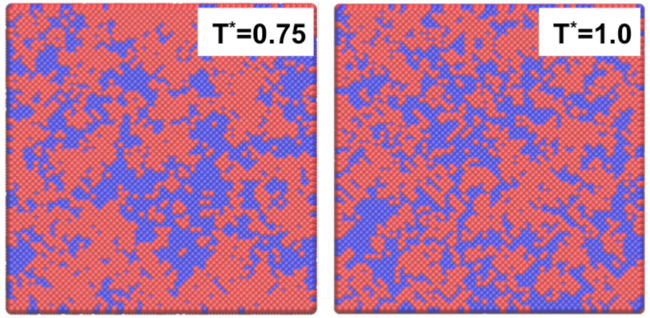


Interacting solid solution model

$$u_i = w_0^p a_i^0 + w_{-1}^p a_i^{-1} + w_1^p a_i^1$$

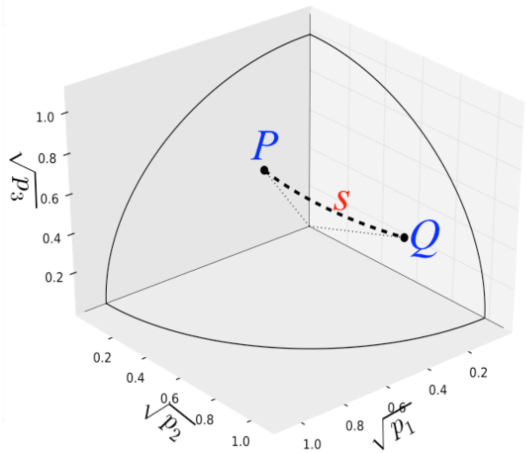
$$a_i^0 = \sum_{NN} \delta^0 ; \quad a_i^{-1} = \sum_{NN} \delta^{-1} ; \quad a_i^1 = \sum_{NN} \delta^1$$





Minimize statistical distance between histograms

$$s = \arccos\left(\sum_{i=1}^{k} \sqrt{p_i}\sqrt{q_i}\right)$$



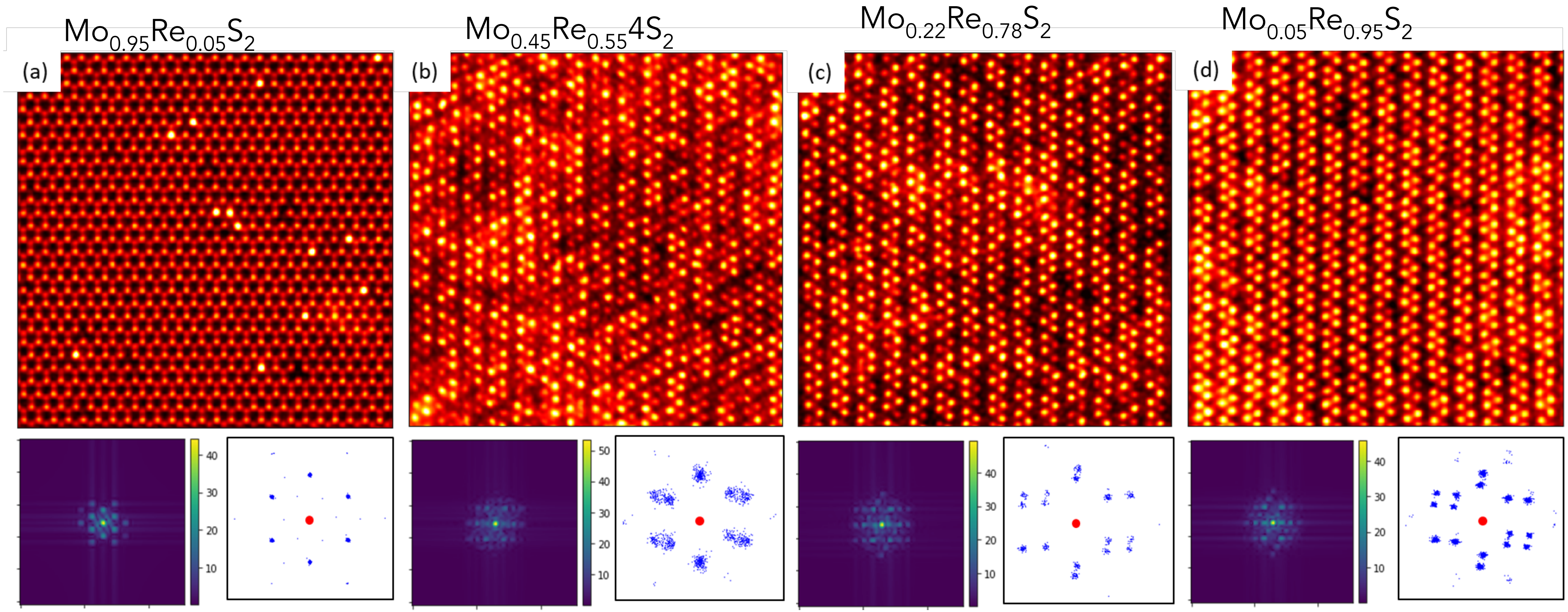
- Utilize all available statistical information in the image
- Using generative model, infer parameters of the model via the experiment.

Vlcek et al., ACS Nano **11**, 10313 (2017)

1) Wootters W.K. *Phys. Rev. B.* **(23)2**, 357-362 (1981).
2) Braunstein S.L., Caves S.M.. *Phys. Rev. Lett.* **72(22)**, 3439-3443 (1994).

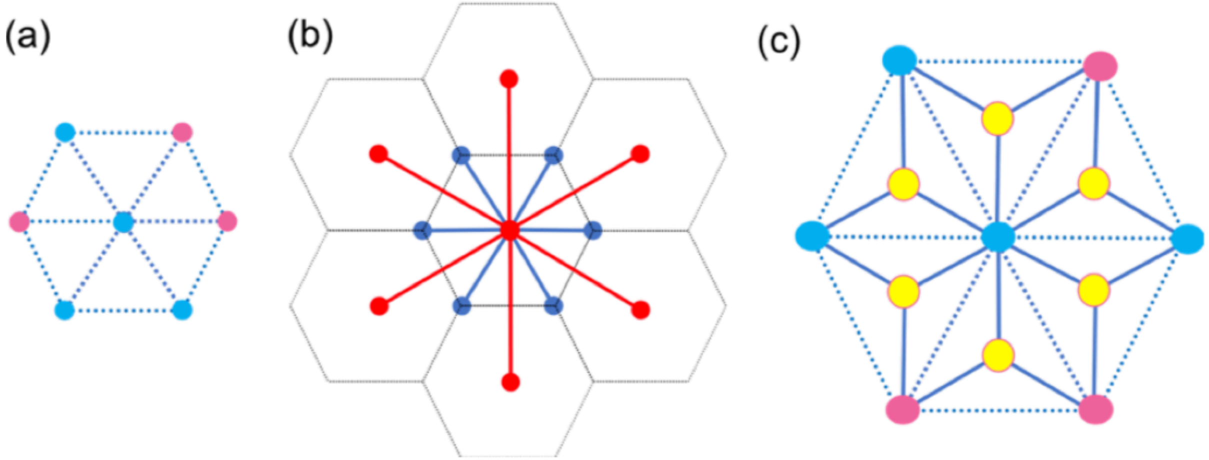
Oak Ridge National Laboratory

Application: MoS_2-ReS_2 system



L. Vlcek, et al., *Appl. Phys. Rev.* 8, 011409 (2021)

Application: MoS_2-ReS_2 system



Cyan – Re
Pink - Mo

Blue – Nearest Neighbor
Red – Next Nearest Neighbor

Yellow: Sulfur

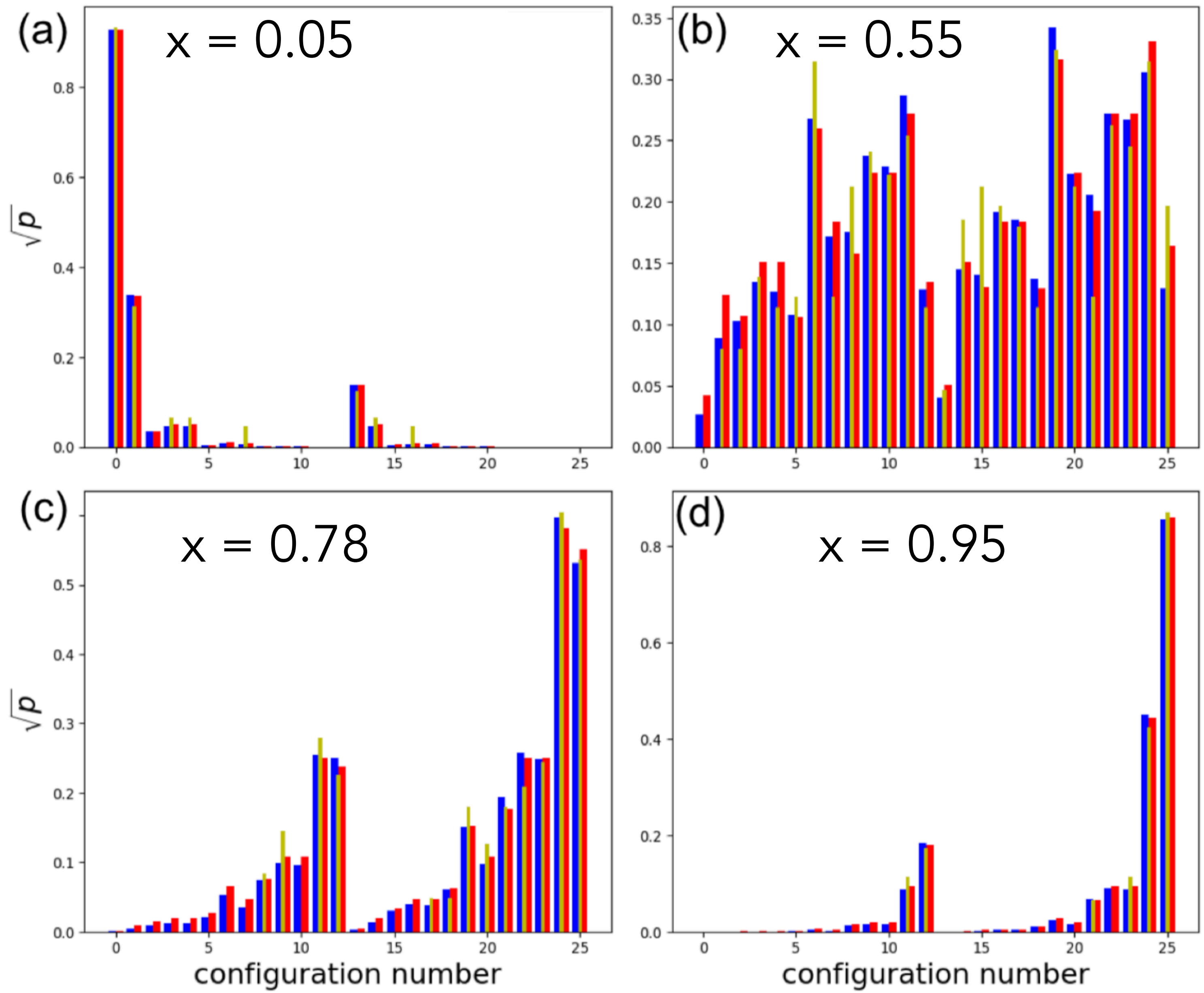
1. Pair-additive Model

$$u_i = w_1 \sum_{\{NN\}} \delta_{MoRe} + w_2 \sum_{\{NNN\}} \delta_{MoRe}$$

2. Many-body Model

$$u_i = w_1 \sum_{\{S\}} \delta_{MoMoRe} + w_2 \sum_{\{S\}} \delta_{MoReRe}$$

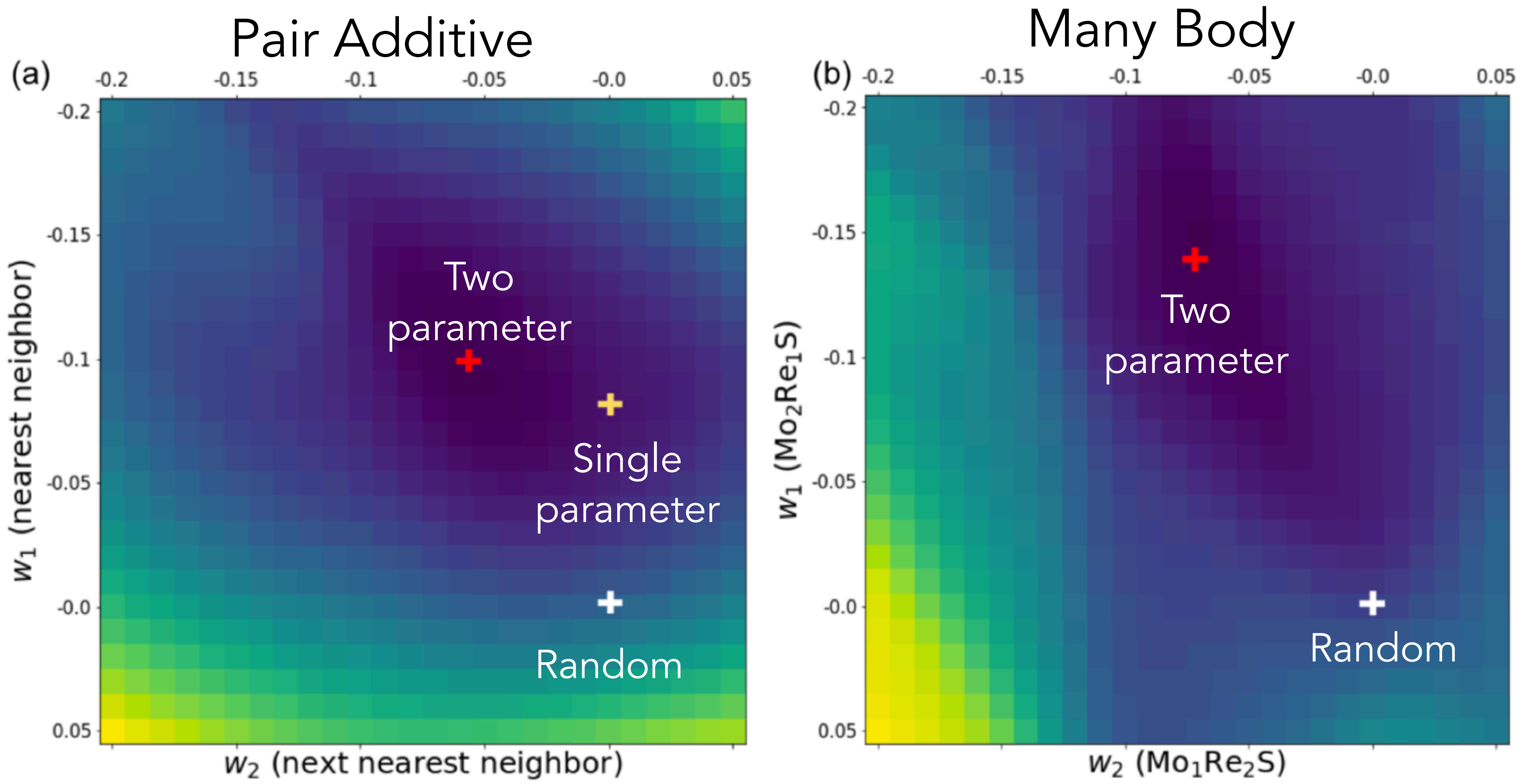
Application: MoS_2-ReS_2 system



Red – Random
Blue – Pairwise Model
Teal - Experiment

1. Random model good at low and high concentrations, but not so great in between.
2. Equilibrium model better, but still not great.

Application: MoS_2-ReS_2 system

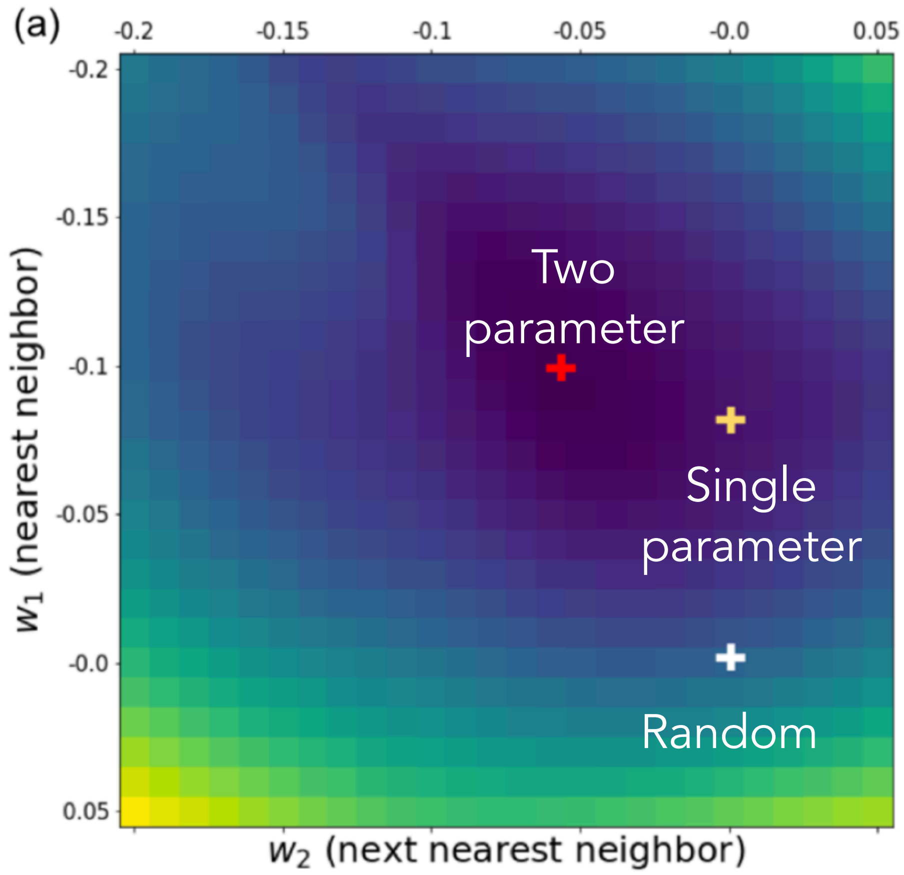


Statistical distance landscapes for two distinct choices of the model

Vlcek et al. npj Comp. Mater. 7, 42 (2021)

Extension: UQ for interactions

Loss landscape (s^2)



- We wish to determine the form of p(**w**|d)
- We can obtain estimates of the mean and variance via bootstrapping:
  - obtain M images;
  - optimize model on a subset
  - Repeat for different subsets
  - Provides uncertainty estimate
- Alternatively, simply use MC sampling to determine p(**w**|d) with log-likelihood given by s^2
- But what about model uncertainty?

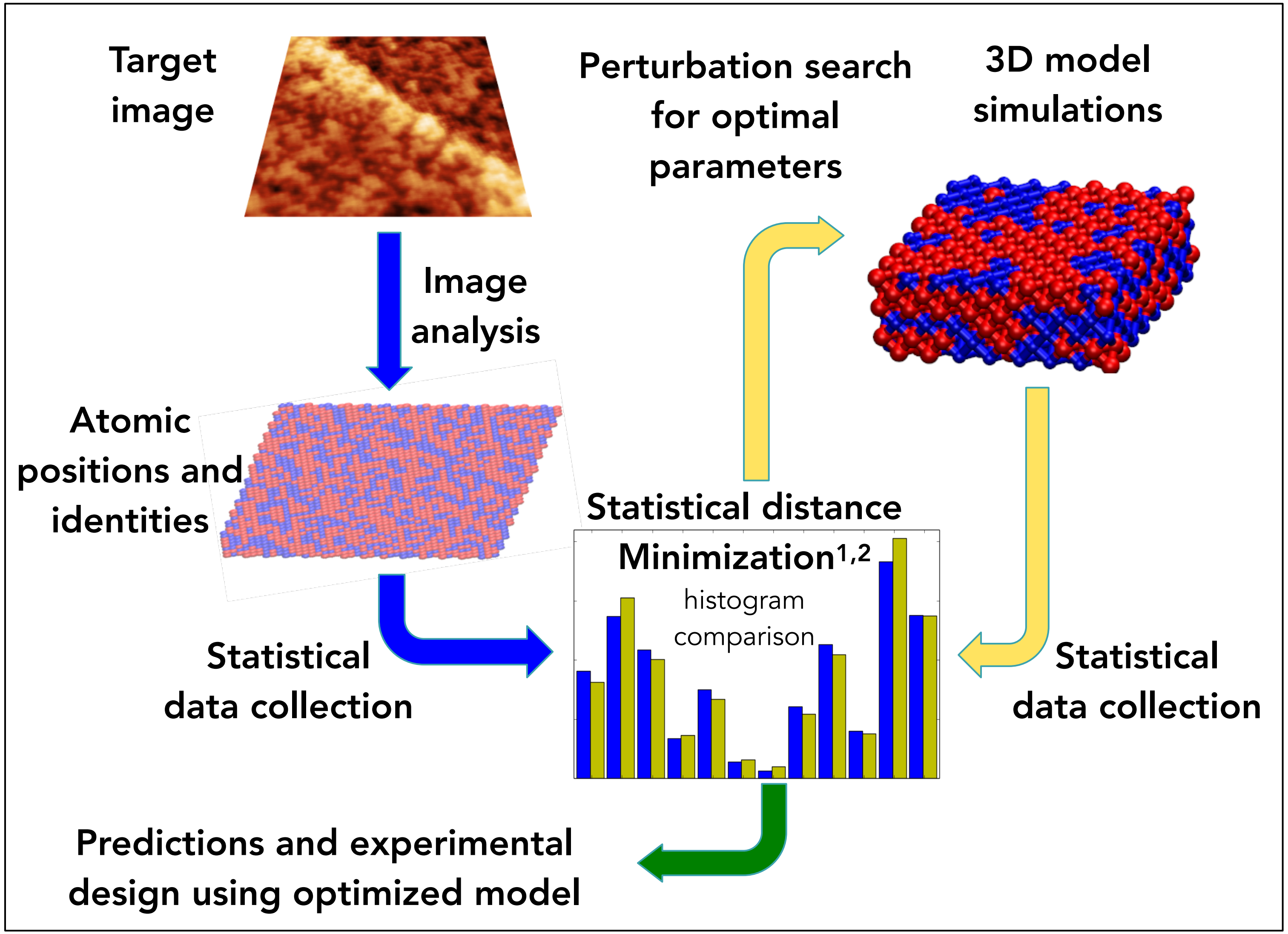
Model selection: Information Criterion

dataset	N	S^2 (R)	PV (R)	S^2 (P1)	PV (P1)	S^2 (P2)	PV (P2)	S^2 (M)	PV (M)
x=0.05	464	0.0062	0.9899	0.0067	0.9829	0.0070	0.9771	0.0073	0.9680
x=0.55	466	0.0325	0.0001	0.0283	0.0009	0.0288	0.0007	0.0284	0.0009
x=0.78	434	0.0298	0.0013	0.0263	0.0069	0.0247	0.0143	0.0275	0.0040
x=0.95	471	0.0124	0.5602	0.0129	0.4976	0.0121	0.5852	0.0121	0.5864
Total	1835	0.0200	0.0015	0.0184	0.0107	0.0180	0.0168	0.0187	0.0079
BIC		73.4		**71.4**		73.6		76.1	

Statistical hypothesis testing for uncertainty

L. Vlcek, S. Yang, Y. Gong, P. Ajayan, W. Zhou, M. F. Chisholm, M. Ziatdinov, R. K. Vasudevan, S. V. Kalinin, [submitted], 2019.

Summary



- We present a framework for analysis of atomically resolved imaging data that captures the physics inherent in the stochastic variations observed
- We combine this with model selection via information criterion approaches
- If applied in real-time, this can help to answer the question of 'how much data is required'